Lecture 12

Recall :

<u>Theorem</u> : Suppose $f = \text{grad } F = \nabla F$.
Then, along any path $c(t)$, $a \le t \le b$,

$$\int \vec{f} \cdot d\vec{c} = F(c(b)) - F(c(a)).$$

This is the fundamental theorem of line integrals.

It essentially says that the line integral of ∇F only depends on the endpoints $c(b)$ and $c(a)$, and not the path itself.

<u>Examples</u> :

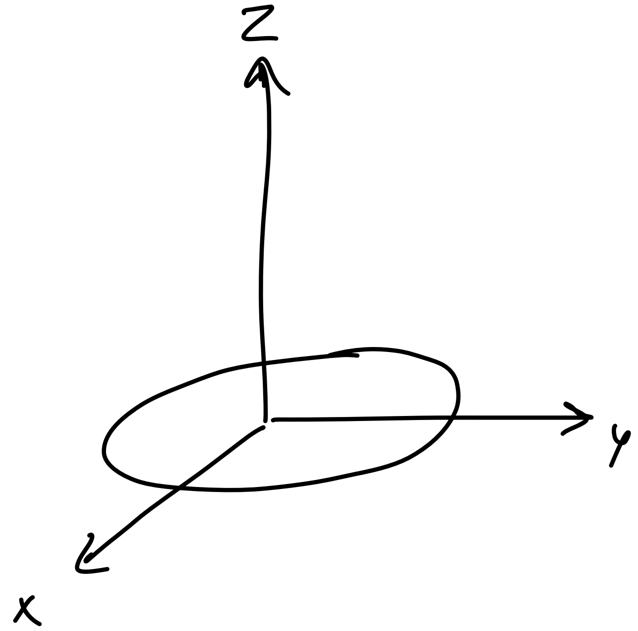
- Suppose $F(x, y, z) = x^2 - y^2 - \sin z$

Integrate ∇F along the semicircular curve joining $(2,0,0)$ and $(-1,1,0)$ in the xy-plane.

By the theorem above,

$$\int \vec{\nabla F} \cdot d\vec{c} = F(2,0,0) - F(-1,1,0)$$
$$= 4.$$

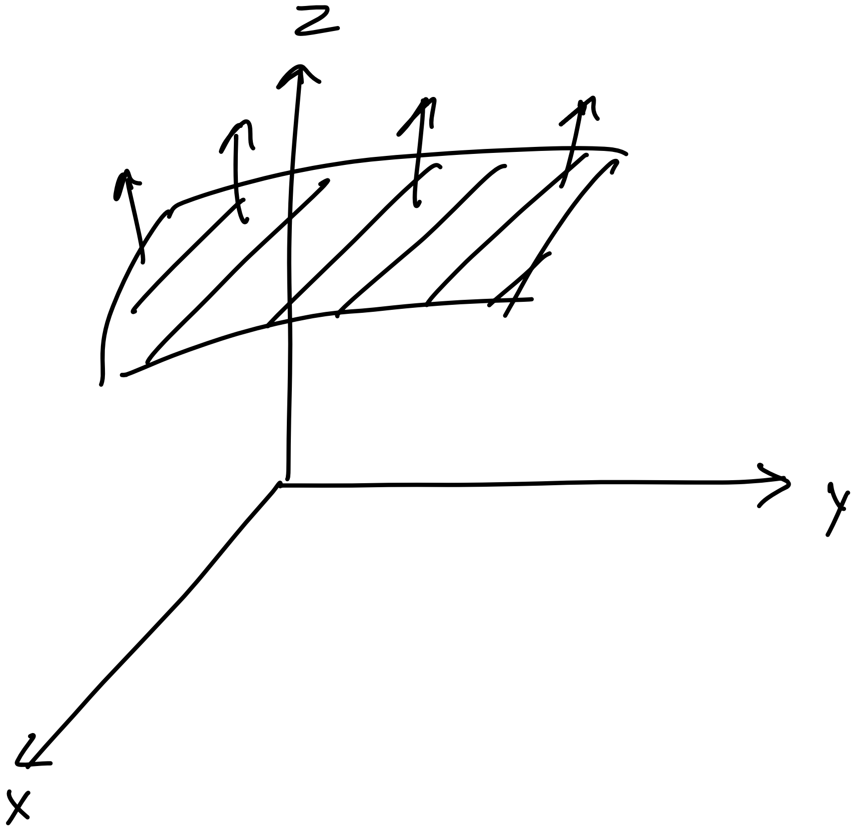
Corallary: What is the integral of ∇F along the unit circle in the xy-plane?



The integral along any closed loop of ∇F is zero.

Parametrization of surfaces

Goal: Integrate scalar/vector-valued functions over 2D-surfaces.



Given vector valued functions, we can compute a surface integral to obtain flux.

The first step towards computing any kind of surface integrals is parametrization.

1) For example, consider the surface

$$x^2 + y^2 + z^2 = 1$$

Recall: In the 1 variable case, we tried to express curves using a single variable t.

For example, $(t^2, t-1, e^t)$ $0 \leq t \leq 2$.

Similarly, for surfaces, we'd ideally want to find parametrizations in terms of 2 variables u and v. For example,

$$(u^2 + v^2,\ u e^{v-u},\ \sin(u+v))$$

$$0 \leq u \leq 1$$

$$1 \leq v \leq 2$$

We'd like to do the same for

$$x^2 + y^2 + z^2 = 1.$$

We can do in 2 ways:

a) $z^2 = 1 - x^2 - y^2$

$z = \pm \sqrt{1-x^2-y^2}$

Hence, our surface is

$(x, y, \sqrt{1-x^2-y^2})$ and

$(x, y, -\sqrt{1-x^2-y^2})$.

$-1 \leq x \leq 1$

$-\sqrt{1-x^2} \leq y \leq \sqrt{1-x^2}$

b) Spherical coordinates.

$x = \sin\phi \cos\theta$

$y = \sin\phi \sin\theta$

$z = \cos\phi$

$0 \leq \phi \leq \pi$

$0 \leq \theta \leq 2\pi$.

2) $(x, y, z) = (u\cos v,\ u\sin v,\ u)$

$0 \leq u \leq 1 \qquad 0 \leq v \leq \pi.$

Find a Cartesian equation for this surface.

$x = u\cos v$

$y = u\sin v$

$z = u$

Then, $x^2 + y^2 = u^2\cos^2 v + u^2\sin^2 v$

$= u^2$

$= z^2$

Hence, the surface is

$$x^2 + y^2 = z^2,$$

which is a cone centred around the z-axis.

3) $(x, y, z) = (u\cos v,\ u\sin v,\ u^2+v^2)$

$0 \leq u \leq 1, \qquad 0 \leq v \leq 2\pi$

As in the previous example,

$$x^2+y^2 = u^2\cos^2 v + u^2\sin^2 v$$
$$= u^2$$
$$= z - v^2$$

$$\Rightarrow v^2 = z - x^2 - y^2$$

Another equation: $\frac{y}{x} = \frac{u\sin v}{u\cos v}$

$$= \tan v.$$

The first equation is equivalent to

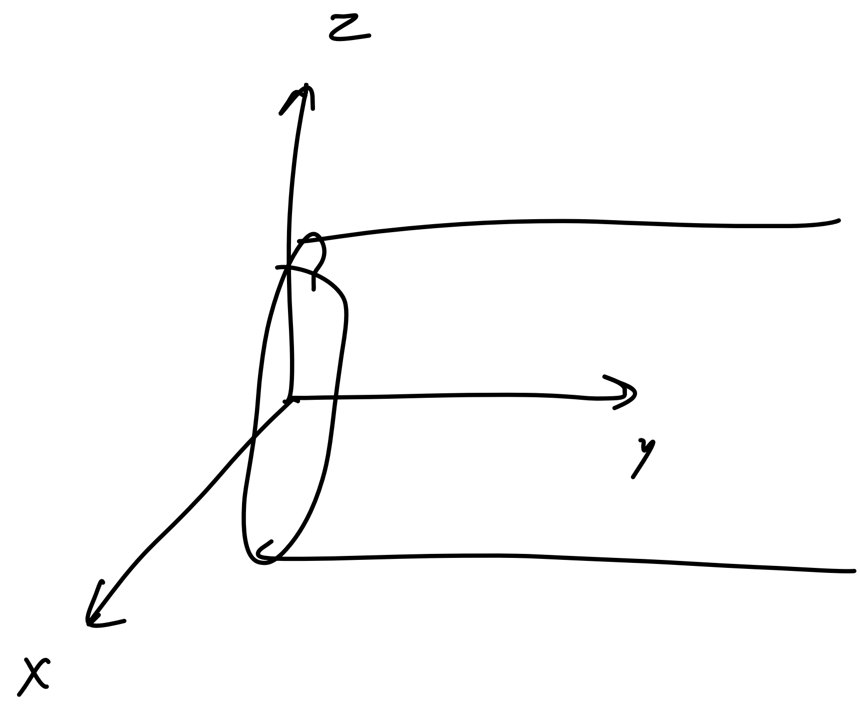
$$v = \sqrt{z-x^2-y^2}$$

So, $\tan\left(\sqrt{z-x^2-y^2}\right) = \frac{y}{x}$

4) Parametrize the surface

$$x^2+z^2=4$$

such that y is non-negative.



Parametrization :

$x = 2\cos u$

$y = v$

$z = 2\sin u$

$0 \le u \le 2\pi$

$0 \le v$.

5) Parametrize the surface

$$x^2 + y^2 - z^2 = 1 .$$

Guess : $x = u\cos v$

$y = u\sin v$

So, the equation becomes

$$u^2\cos^2 v + u^2\sin^2 v - z^2 = 1$$
$$\Rightarrow \quad u^2 - z^2 = 1$$

Recall : $\sec^2\theta - \tan^2\theta = 1$.

So, we'd want
$$u = \sec w$$
$$z = \tan w.$$

So, the parametrization is :

$$x = u\cos v = \sec w \cos v$$
$$y = u\sin v = \sec w \sin v$$
$$z = \tan w$$

$$0 \le v \le 2\pi.$$
$$0 \le w \le 2\pi.$$

7) Parametrize the plane
$$x + 2y + 3z = 6$$

Solve for y:

$$y = \frac{6 - x - 3z}{2}$$

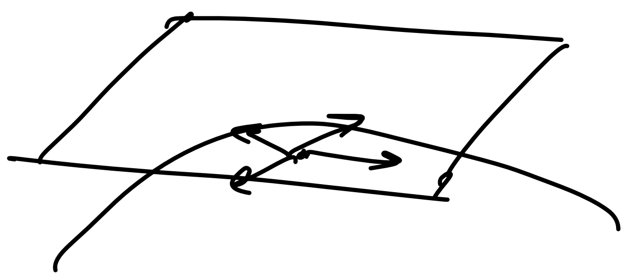
Parametrization:

$$x = x$$

$$y = \frac{6 - x - 3z}{2}$$

$$z = z$$

Given a parametrized surface, we would want to find the equation of the tangent plane.



Strategy:

- Find 2 tangent vectors at the point.

- Take their cross product to find a normal vector.
- Find the equation of the plane.

Given a parametrized surface:

$$(f(u,v),\ g(u,v),\ h(u,v))$$
$$\quad\quad\quad\quad \shortparallel$$
$$(x, y, z).$$

How do we find equations of tangent vectors?

We can find derivatives in the u and v-directions to get tangent vectors T_u and T_v.

Example:

- $(x, y, z) = (2u,\ u^2 + 2v,\ v^2)$

We can find 2 tangent vectors:

$$T_u = (2, 2u, 0)$$
$$T_v = (0, 2, 2v)$$

A normal vector can then be found as a cross product

$$T_u \times T_v$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2u & 0 \\ 0 & 2 & 2v \end{vmatrix}$$

$$= 4uv\,\hat{i} - 4v\,\hat{j} + 4\,\hat{k}.$$